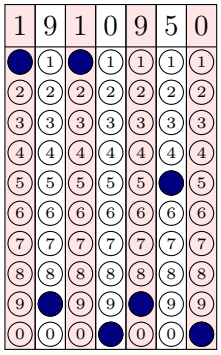
TEST VARIANTI RAQAMI: **1910950**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



1. Piramida balandligining o‘rtasidan asosga parallel tekislik o‘tkazilgan va hosil bo‘lgan kesim yuzi 27 ga teng. Agar berilgan piramidaning balandligi 10 ga teng bo‘lsa, uning hajmini toping.

A) *340* B) *400* C) *360* D) *320*

2. Uchlari $A(3;0)$, $B(0;2)$ va $C(0;0)$ nuqtalarda bo‘lgan uchburchakning CM bissektrisasi bo‘lsa, M nuqtaning koordinatalarini toping.

A) $\left(\frac{6}{5};\frac{6}{5}\right)$ B) $\left(\frac{3}{4};\frac{3}{4}\right)$ C) $\left(\frac{4}{3};\frac{4}{3}\right)$
D) $\left(\frac{5}{6};\frac{5}{6}\right)$

3. Ifodani soddalashtiring:
$\left(\frac{\sqrt[4]{m}-4}{\sqrt[4]{m}+4}-\frac{\sqrt[4]{m}+4}{\sqrt[4]{m}-4}\right)^{-1}\cdot\left(\frac{4\sqrt[4]{m}}{\sqrt[4]{m}-4}\right)$

A) $-\frac{\sqrt[4]{m}+4}{4}$ B) $-\sqrt[4]{m}-4$ C) $\frac{\sqrt[4]{m}+4}{4}$
D) $-4\,(\sqrt[4]{m}+4)$

4. $-1,25$ soniga qarama-qarshi bo‘lgan sonning teskarisi $0,1$ dan qanchaga katta?

A) *0,3* B) *1,15* C) *0,4* D) *0,7*

5. $y=-x^2+bx+c$ kvadrat funksiyaning eng katta qiymati -2 ga teng va unga $x=2$ nuqtada erishadi. bc ni toping.

A) *24* B) -24 C) *18* D) -18

6. Birinchi moddiy nuqta $S_1(t)=t^3-3t^2+6t+2$ [m], ikkinchisi $S_2(t)=\frac{1}{3}t^3+t^2-3t+1$ [m] qonuniyat bo‘yicha harakatlanmoqda. Ularning tezlanishlari teng bo‘lgan vaqtdagi birinchisining tezligini (m/s) toping. (t – vaqt, s)

A) *9,5* B) *4* C) *6* D) *8*

7. Tengsizlikni yeching:
$\sqrt[3]{(x-2)^3}+\sqrt[4]{(x-2)^4}\le 2-x$.

A) $\{2\}$ B) $\varnothing$ C) $(-\infty;2]$ D) $[2;\infty)$

8. $\frac{1}{\sqrt{x^3+2}+3}+\frac{2}{\sqrt{x^3+2}+7}=\frac{1}{2}$ tenglama nechta haqiqiy ildizga ega?

A) *2* B) *4* C) *1* D) *0*

9. 3 ta turli lavozimga nomzodlari ko‘rsatilgan 5 kishidan 3 kishini necha xil usul bilan saylash mumkin?

A) *60* B) *70* C) *56* D) *64*

10. Agar a va b natural sonlar yig‘indisi 7 ga qoldiqsiz bo‘linsa, $37a+9b$ ni 7 ga bo‘lgandagi qoldiqni toping.

A) *0* B) *6* C) *1* D) *2*

11. a va b natural sonlar uchun $a+\frac{b}{3}=10$ bo‘lsa, u holda ab ifodaning eng katta qiymatini toping.

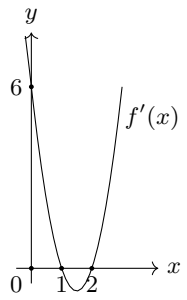
A) *63* B) *72* C) *54* D) *75*

12. Sharga tashqi chizilgan kesik konus asosining radiuslari 3 va 6 ga teng. Kesik konus to‘la sirtining shar sirtiga nisbatini toping.

A) $\frac{7}{4}$ B) $\frac{14}{9}$ C) $\frac{14}{3}$ D) $\frac{7}{6}$

© Davlat test markazi, 2019

13. Rasmda $f'(x)$ kvadrat funksiyaning grafigi tasvirlangan. Uning boshlang'ich funksiyasi uchun $f(0)=0$ bo'lsa, u holda $f(2)$ ni toping.



A) *1* B) *2* C) *0* D) *3*

14. ABC uchburchakning BD medianasidagi E va F nuqtalar (E nuqta B uchiga yaqin joylashgan) medianani teng uchta qismga bo'ladi. Agar ABC uchburchakning yuzi 36 ga teng bo'lsa, AEB uchburchakning yuzini toping.

A) *6* B) *4* C) *12* D) *9*

15. Grafigi $A(-2;11)$ nuqtadan o'tuvchi $y=kx+5$ funksiya abssissalar o'qini qaysi nuqtada kesib o'tadi?

A) $\left(-1\frac{2}{3};0\right)$ B) $\left(1\frac{2}{3};0\right)$ C) $\left(1\frac{1}{2};0\right)$ D) $\left(0;1\frac{1}{2}\right)$

16. -3; 6; 8 va x sonlarining o'rta arifmetigi y ning $\frac{1}{3}$ qismiga teng. Agar $3x-2y=15$ bo'lsa, y ning qiymatini toping.

A) *32* B) *24* C) *18* D) *28*

17. A va B to'plamlarning elementlari mos ravishda 24 va 36 sonlarining natural bo'luvchilaridan iborat bo'lsa, $A\cap B$ to'plamning elementlari sonini aniqlang.

A) *6* B) *4* C) *12* D) *8*

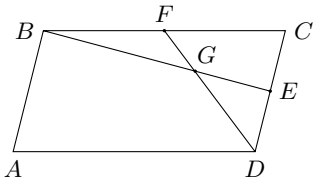
18. Uzunligi 80 metr bo'lgan sim uzunliklari 5:7:13 nisbatda bo'lingan. Hosil bo'lgan simlardan eng yengilining uzunligini (m) toping.

A) *17* B) *16* C) *16,4* D) *16,6*

19. Agar $0<\alpha,\ \beta<\frac{\pi}{2}$ lar uchun $tg\alpha=\frac{1}{2}$ va $\sin\beta=\frac{15}{17}$ bo'lsa, $\sin 2\alpha+tg\frac{\beta}{2}$ ni hisoblang.

A) $\frac{7}{5}$ B) $\frac{2}{5}$ C) $\frac{5}{7}$ D) $\frac{5}{2}$

20. Rasmda $ABCD$ parallelogramm tasvirlangan. G nuqta BE va DF kesmalarning kesishish nuqtasi. Agar $BF=FC$ va $CE=ED$ bo'lsa, $\frac{S_{ABCD}}{S_{ABGD}}$ ni toping.



A) 1 B) $\frac{3}{2}$ C) 2 D) $\frac{5}{3}$

21. Hisoblang: $\left(\sqrt{11-6\sqrt{2}}-2\right)^2+2\sqrt{2}$.

A) 4 B) 3 C) $-2\sqrt{2}$ D) 1

22. Ikkita to'g'ri chiziq kesishganidan hosil bo'lgan burchaklardan biri ikkinchisidan 36° ga katta bo'lsa, ularning nisbatini toping.

A) *4:3* B) *5:4* C) *3:2* D) *6:5*

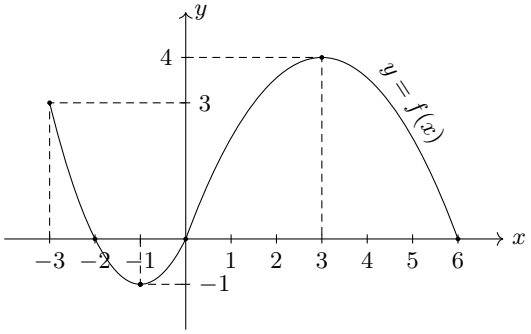
23. $y=3^{x-3}+12$ funksiyaning qiymatlar sohasini toping.

A) $(12;\infty)$ B) $(15;\infty)$ C) $(-\infty;\infty)$ D) $[12;\infty)$

24. Arifmetik progressiyani tashkil etuvchi $n+3; n+9; n+15; ...; n+123$ ketma-ketlikning o'n birinchi hadi 65 ga teng bo'lsa, bu ketma-ketlikning uchinchi hadini toping.

A) *18* B) *16* C) *17* D) *14*

25. Rasmda $y=f(x)$ funksiyaning grafigi tasvirlangan. $f'(x)\cdot f(x)\ge 0$ tengsizlikning $(0;\ 6)$ oraliqdagi yechimlarini toping.



A) [3; 6) B) $\varnothing$ C) (0; 3] D) (0; 6)

26. $\frac{x^2+9}{4-x^2}=\frac{\sqrt{9+x^2}}{\sqrt{4-x^2}}$ tenglama nechta haqiqiy ildizga ega?

A) *2* B) *haqiqiy ildizga ega emas* C) *1* D) *4*

© Davlat test markazi, 2019

27. Agar $x = 2n + 1$ (n – natural son) bo‘lsa, $\dfrac{(-1)^{x+1} + (-1)^{2x}}{(-1)^{x}}$ ifodaning qiymatini aniqlang.

A) -2 B) *0* C) *2* D) *−2 yoki 0*

28. $(3 - a)(a + 4) - a(-a - 6)$ ifodaning $a = 2\dfrac{2}{5}$ bo‘lgandagi qiymatini toping.

A) *24* B) $10\dfrac{2}{5}$ C) *21* D) $8\dfrac{3}{5}$

29. $\sin 4x = \sin 3x$ tenglamaning eng kichik musbat yechimini toping.

A) $\dfrac{6\pi}{7}$ B) $\dfrac{2\pi}{7}$ C) $\dfrac{\pi}{7}$ D) $\dfrac{4\pi}{7}$

30. $\begin{cases} \log_2 (x-4)^2 \le 2 \\ (x-1)^2 > 4 \end{cases}$ tengsizliklar sistemasi nechta butun yechimga ega?

A) *cheksiz ko‘p* B) *3 ta*
C) *butun yechimga ega emas* D) *2 ta*

Test varianti “Test topshiriqlari to‘plami 2019” asosida shakllantirilgan:

1 – 109-bet 23	7 – 69-bet 25	13 – 87-bet 20	19 – 44-bet 93	25 – 81-bet 39
2 – 116-bet 20	8 – 66-bet 74	14 – 95-bet 41	20 – 103-bet 100	26 – 60-bet 9
3 – 50-bet 49	9 – 125-bet 21	15 – 73-bet 37	21 – 28-bet 56	27 – 24-bet 22
4 – 12-bet 19	10 – 7-bet 6	16 – 20-bet 40	22 – 90-bet 5	28 – 48-bet 29
5 – 76-bet 72	11 – 14-bet 41	17 – 119-bet 6	23 – 55-bet 31	29 – 36-bet 22
6 – 80-bet 32	12 – 113-bet 57	18 – 17-bet 9	24 – 30-bet 11	30 – 56-bet 13

© Davlat test markazi, 2019